Recitation 3: Big-O and Computational Complexity

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1 Big-O

True or False? Give a brief justification for each.

1. $n \log n \in O(n)$
2. $10^9 n \in \Theta(n)$
3. $\log(n^4) = O(\log n)$
4. $n^6 \cdot 2^{-n} \in O(1)$
5. $3^n \in O(4^n)$
6. $2^{\sqrt{n}} \in O(n^3)$
7. $2^n \in \Theta(n^{1{,}000{,}000})$
8. $n^k \in O(k^n)$ for $k \geq 2$

Formally prove the following statements:

1. $3n^2 + 7n \log n + 12n + 5 \in O(n^2)$
2. $n^2 + 3/n \in O(n^2)$
3. $n^2 + 3n \log n \in O(n^2)$
4. $n \log n \in O(n^{1+\epsilon})$ for every $\epsilon > 0$

2 Computational Complexity

1. Analyze the running time of the following program:

```
v = 0
for i = 1 to n do
    for j = 1 to n do
        v = i · j^2
    end for
end for
```

2. Analyze the running time of the following program (you may assume that A is already sorted):

```
i, j = 0, length(A) − 1
while i < j do
    if A[i] ! = A[j] then
        return False
    end if
    i = i + 1
    j = j − 1
end while
return True
```

3. Suppose the function $F(i, j)$ takes $\Theta(i)$ time to execute. Analyze the runtime of the following program, which has input n:

```
for i = 1 to n do
    for j = i to n do
        Call F(i, j)
    end for
end for
```

   Hint: $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

4. Given an array of integers *nums* and an integer target *num*, return indices of the two numbers such that they add up to *target*. You may assume that each input would have exactly one solution, and you may not use the same element twice. What is the runtime of your solution? Can you come up with a solution with a runtime of $\Theta(n)$?